

Comptia A+ Assignment

[A+ - Understanding and maintenance of Networks]

Network Cable Types and Tools

Types of cables (Ethernet, fiber optics) and tools (RJ45 crimper). Create a straight-through Ethernet cable.

Task

- ❖ **1. List 3 key differences between Ethernet cables and fiber optic cables, and give one real-world example where each type would be used (for example, home WiFi router, data center, college campus backbone).**

➤ Ethernet Cables vs Fiber Optic Cables:

- Ethernet and fiber optic cables are both used for network communication, but they differ in their transmission method, speed/distance capabilities, and installation requirements.

Difference	Ethernet Cable	Fiber Optic Cable
Transmission Method	Transmits data using <u>electrical signals</u> through copper wires.	Transmits data using <u>light signals</u> through optical fibers.

Distance & Speed	Generally suitable for shorter network connections and common LAN installations.	Supports <u>much longer distances and very high bandwidth</u> , making it suitable for large networks.
Cost & Installation	Usually cheaper, easier to install, and commonly used for everyday networking.	Usually more expensive and requires specialized equipment and installation techniques.

➤ Real-World Examples:

- **Ethernet Cable:** Commonly used to connect a home Wi-Fi router to a desktop PC, gaming PC, printer, or network switch.
- **Fiber Optic Cable:** Commonly used as a college campus backbone or data-center connection where high bandwidth and long-distance communication are required.

❖ **2. Draw and label a diagram showing the internal wiring of a straight-through Ethernet cable (use T568B wiring standard), indicating the color of each wire and its pin number.**

➤ Straight-Through Ethernet Cable – T568B:

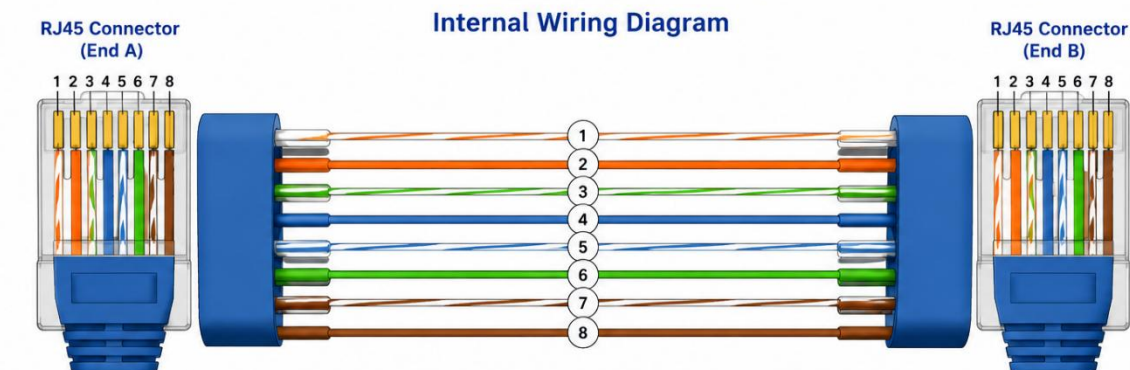
- A straight-through Ethernet cable uses the T568B wiring standard on both RJ45 ends. The eight wires are arranged in a fixed color and pin

order. Since both ends use the same T568B arrangement, it is called a straight-through cable and is commonly used to connect devices such as a computer to a switch or a router to a switch.

➤ T568B Pin Order:

- White/Orange
- Orange
- White/Green
- Blue
- White/Blue
- Green
- White/Brown
- Brown

STRAIGHT-THROUGH ETHERNET CABLE (T568B STANDARD)



T568B WIRING STANDARD			
Pin No.	Wire Color	Pair No.	Function
1	White / Orange	2	Transmit Data + (TX+)
2	Orange	2	Transmit Data - (TX-)
3	White / Green	3	Receive Data + (RX+)
4	Blue	1	Unused
5	White / Blue	1	Unused
6	Green	3	Receive Data - (RX-)
7	White / Brown	4	Unused
8	Brown	4	Unused

STRAIGHT-THROUGH CABLE

In a straight-through cable, both ends use the same T568B wiring standard.



REAL-WORLD USE



❖ **3. Watch a YouTube video demonstrating how to use an RJ45 crimper, then write a step-by-step guide (in your own words) on how to attach an RJ45 connector to a CAT6 cable.**

- Steps to Attach an RJ45 Connector to a CAT6 Cable:
- **Tools Required:** CAT6 cable, RJ45 connector, RJ45 crimping tool, and cable tester.



1. **Cut the Cable** – Cut the CAT6 cable to the required length.
2. **Strip the Cable** – Remove a small part of the outer cover carefully.
3. **Separate the Wires** – Separate and straighten all eight wires.
4. **Arrange the Wires** – Arrange the wires in the **T568B color order**: White/Orange, Orange, White/Green, Blue, White/Blue, Green, White/Brown, Brown.
5. **Trim the Wires** – Make all eight wires equal in length.
6. **Insert the Wires** – Insert the wires into the RJ45 connector in the correct order and make sure they reach the end.
7. **Crimp the Connector** – Place the RJ45 connector in the crimping tool and press it firmly to secure the wires.
8. **Test the Cable** – Use a cable tester to check that all pins are connected correctly.

9. Source video: [CAT-6 Cable Crimping Tutorial – RJ45 T568B](#)

➤ Result:

- The RJ45 connector was successfully attached to the CAT6 cable using the T568B standard, and the cable was tested for proper connectivity.



Cut the cable to the required length.



Strip about 1–1.5 inches of the outer jacket.



Separate and straighten the 8 wires.



Arrange the wires in T568B order.



Trim the wires evenly.



Insert wires into RJ45 connector (pin 1 to 8).



Crimp the connector using RJ45 crimper.



Test the cable. Pins 1 to 8 should light up in order.

❖ **4. Imagine you are setting up a gaming café with 10 PCs for LAN gaming. Decide which cable type (Ethernet or fiber optics) and which tool (RJ45 crimper or fiber cleaver) you would use, and explain your choices in 3-4 sentences. Consider speed, distance, and budget for your answer.**

➤ Gaming Café Network Setup:

- For a gaming café with 10 PCs, I would choose CAT6 Ethernet cables because they provide high speed and are reliable for short-distance LAN gaming. I would use an RJ45 crimper to attach RJ45 connectors to the Ethernet cables. Ethernet is more cost-effective and easier to install than fiber optic cables for a small café network. Fiber optic cables and a fiber cleaver would be more suitable for long-distance connections or high-capacity network backbones.